Robust Angle-Uncertainty-Aware Joint ISAC Waveform Design and Simulation

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***Abstract*—This report studies a robust joint integrated sensing and communication (ISAC) waveform for multi-user communication and multi-target sensing. Communication quality is enforced by constructive-interference (CI) constraints, while sensing quality is measured by beampattern matching mean-square error (MSE). Target angles are modeled by an ellipsoidal uncertainty set. The full formulation is written as a robust multi-objective problem with weighted Tchebycheff scalarization, then relaxed by semidefinite relaxation (SDR). The MATLAB simulation uses a simplified sensing-oriented joint ISAC problem with fixed CI thresholds. Results show that the robust design slightly sacrifices nominal rate and nominal MSE, but reduces high-risk and worst-case sensing errors under angle uncertainty.**

***Index Terms*—ISAC, MIMO, constructive interference, robust optimization, angle uncertainty, beampattern matching, SDR.**

I. Introduction

JOINT ISAC waveform design must support communication users and radar sensing targets using the same transmit signal. A nominal design works well when target angles are exact, but its beampattern can degrade when the real target angles shift. This report therefore considers an angle-uncertainty-aware robust design. The goal is to keep the CI communication constraints feasible and reduce sensing MSE under target angle errors.

II. System Model and Formulation

A. System Assumptions

A dual-functional radar-communication base station (BS) has N_t transmit antennas and N_r receive antennas. Both arrays are half-wavelength ULAs. The BS serves K single-antenna users and senses N point targets:

$$\mathcal{K} = \{1, 2, \ldots, K\}, \qquad \mathcal{N} = \{1, 2, \ldots, N\}. \tag{1}$$

The communication link is downlink BS-to-user. The sensing link is BS-to-target-to-BS.

B. Transmit Signal and Covariance

Let the communication symbol vector and precoding matrix be

$$\mathbf{s} = [s_1, s_2, \ldots, s_K]^T \in \mathbb{C}^{K\times 1}, \tag{2}$$

$$\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_K] \in \mathbb{C}^{N_t\times K}. \tag{3}$$

The symbol-level transmit waveform is

$$\mathbf{x} = \mathbf{W}\mathbf{s} = \sum_{k=1}^{K} \mathbf{w}_k s_k. \tag{4}$$

For a deterministic waveform, define

$$\mathbf{R} = \mathbf{x}\mathbf{x}^H. \tag{5}$$

Before relaxation, $\mathbf{R} \succeq 0$ and $\mathrm{rank}(\mathbf{R}) = 1$. The power constraint is

$$\mathrm{tr}(\mathbf{R}) = \|\mathbf{x}\|_2^2 \le P_{\max}. \tag{6}$$

C. Communication Model

For user k, the received signal is

$$y_k = \mathbf{h}_k^H \mathbf{x} + n_k, \tag{7}$$

where $\mathbf{h}_k \in \mathbb{C}^{N_t\times 1}$ and $n_k \sim \mathcal{CN}(0, \sigma^2)$. The received SNR is

$$\gamma_k = \frac{|\mathbf{h}_k^H \mathbf{x}|^2}{\sigma^2}. \tag{8}$$

The achievable rate and sum rate are

$$R_k = B \log_2(1 + \gamma_k), \qquad R_{\mathrm{sum}}(\mathbf{x}) = \sum_{k=1}^{K} R_k. \tag{9}$$

The communication objective in minimization form is

$$f_C(\mathbf{x}) = -R_{\mathrm{sum}}(\mathbf{x}). \tag{10}$$

D. Constructive Interference Constraints

For symbol-level precoding, the noiseless received signal of user k is

$$\widehat{y}_k = \mathbf{h}_k^H \mathbf{x} = \mathbf{h}_k^H \mathbf{w}_k s_k + \sum_{i\ne k} \mathbf{h}_k^H \mathbf{w}_i s_i. \tag{11}$$

Rotate it by the intended symbol:

$$u_k = \widehat{y}_k s_k^* = \mathbf{h}_k^H \mathbf{x} s_k^*. \tag{12}$$

Substituting $\mathbf{x} = \sum_{i=1}^{K} \mathbf{w}_i s_i$ gives

$$u_k = \mathbf{h}_k^H \mathbf{w}_k |s_k|^2 + \sum_{i\ne k} \mathbf{h}_k^H \mathbf{w}_i s_i s_k^*. \tag{13}$$

For PSK symbols, $|s_k| = 1$. The rotation aligns the desired symbol with the positive real axis, so the CI region can be described by the real and imaginary parts of u_k. For M-PSK, the half decision angle is

$$\phi = \frac{\pi}{M}. \tag{14}$$

The CI region with SNR threshold Γ_k is

$$\mathrm{Re}(u_k) \ge \sqrt{\Gamma_k \sigma^2}, \tag{15}$$

$$|\,\mathrm{Im}(u_k)| \le \left(\mathrm{Re}(u_k) - \sqrt{\Gamma_k \sigma^2}\right) \tan\phi. \tag{16}$$

Equivalently, for all $k \in \mathcal{K}$,

$$\operatorname{Im}(\mathbf{h}_k^H \mathbf{x} s_k^*) \leq \left[\operatorname{Re}(\mathbf{h}_k^H \mathbf{x} s_k^*) - \sqrt{\Gamma_k \sigma^2}\right] \tan\phi, \tag{17}$$

$$-\operatorname{Im}(\mathbf{h}_k^H \mathbf{x} s_k^*) \leq \left[\operatorname{Re}(\mathbf{h}_k^H \mathbf{x} s_k^*) - \sqrt{\Gamma_k \sigma^2}\right] \tan\phi. \tag{18}$$

These linear inequalities keep each symbol in the correct PSK decision region.

E. Sensing Model and Beampattern

The transmit and receive steering vectors are

$$\mathbf{a}_t(\theta) = \frac{1}{\sqrt{N_t}}[1, e^{-j\pi \sin\theta}, \ldots, e^{-j\pi(N_t-1)\sin\theta}]^T, \tag{19}$$

$$\mathbf{a}_r(\theta) = \frac{1}{\sqrt{N_r}}[1, e^{-j\pi \sin\theta}, \ldots, e^{-j\pi(N_r-1)\sin\theta}]^T. \tag{20}$$

For target n with angle θ_n and reflection coefficient β_n, the received sensing signal is

$$\mathbf{y}_r = \sum_{n=1}^{N} \beta_n \mathbf{a}_r(\theta_n) \mathbf{a}_t^T(\theta_n) \mathbf{x} + \mathbf{z}. \tag{21}$$

Only the transmit beampattern is needed for optimization:

$$G(\theta) = \mathbf{a}_t^H(\theta) \mathbf{R} \mathbf{a}_t(\theta). \tag{22}$$

For grid point θ_l, define

$$\mathbf{A}_l = \mathbf{a}_t(\theta_l) \mathbf{a}_t^H(\theta_l), \tag{23}$$

so that

$$G(\theta_l) = \operatorname{tr}(\mathbf{A}_l \mathbf{R}). \tag{24}$$

F. Ellipsoidal Angle Uncertainty

The true target angle is

$$\theta_n = \widehat{\theta}_n + \delta_n. \tag{25}$$

Collect the errors as

$$\boldsymbol{\delta} = [\delta_1, \delta_2, \ldots, \delta_N]^T. \tag{26}$$

The ellipsoidal uncertainty set is

$$\boldsymbol{\delta} = \mathbf{P}\mathbf{u}, \qquad \|\mathbf{u}\|_2 \leq 1, \tag{27}$$

$$\mathcal{U} = \{\boldsymbol{\delta} \mid \boldsymbol{\delta} = \mathbf{P}\mathbf{u},\ \|\mathbf{u}\|_2 \leq 1\}. \tag{28}$$

For diagonal uncertainty,

$$\mathbf{P} = \operatorname{diag}(\epsilon_1, \epsilon_2, \ldots, \epsilon_N), \tag{29}$$

$$\theta_n(\mathbf{u}) = \widehat{\theta}_n + \epsilon_n u_n, \qquad \|\mathbf{u}\|_2 \leq 1. \tag{30}$$

The ellipsoid limits total uncertainty energy, so all targets do not necessarily reach their maximum error together.

G. Desired Beampattern and Sensing MSE

Let $\Theta = \{\theta_1, \theta_2, \ldots, \theta_L\}$ be the angular grid and Δ_θ be the half beam width. For an error $\boldsymbol{\delta}$, the desired beampattern is

$$d_l(\boldsymbol{\delta}) = \begin{cases} 1, & \exists n \in \mathcal{N} : |\theta_l - (\widehat{\theta}_n + \delta_n)| \leq \Delta_\theta, \\ 0, & \text{otherwise.} \end{cases} \tag{31}$$

For any $\boldsymbol{\delta} \in \mathcal{U}$, define

$$M_s(\alpha_s, \mathbf{R}; \boldsymbol{\delta}) = \frac{1}{L} \sum_{l=1}^{L} \left|\operatorname{tr}(\mathbf{A}_l \mathbf{R}) - \alpha_s d_l(\boldsymbol{\delta})\right|^2, \tag{32}$$

where α_s scales the fixed desired pattern and compensates for magnitude mismatch. The worst-case robust sensing objective is

$$f_S(\alpha_s, \mathbf{R}) = \max_{\boldsymbol{\delta} \in \mathcal{U}} M_s(\alpha_s, \mathbf{R}; \boldsymbol{\delta}). \tag{33}$$

With epigraph variable ζ,

$$M_s(\alpha_s, \mathbf{R}; \boldsymbol{\delta}) \leq \zeta, \qquad \forall \boldsymbol{\delta} \in \mathcal{U}, \tag{34}$$

and the sensing objective is $\min_{\alpha_s, \mathbf{R}} \zeta$.

H. Robust Multi-Objective Problem

The two objectives are

$$f_C(\mathbf{x}) = -R_{\text{sum}}(\mathbf{x}), \qquad f_S(\alpha_s, \mathbf{R}) = \zeta. \tag{35}$$

The robust multi-objective problem is

$$\min_{\mathbf{x}, \alpha_s, \mathbf{R}} \begin{bmatrix} f_C(\mathbf{x}) \\ f_S(\alpha_s, \mathbf{R}) \end{bmatrix} \tag{36}$$

subject to CI, robust MSE, power, and covariance constraints.

Let f_C^* and f_S^* be the communication-only and sensing-only ideal values. The normalized objectives are

$$\bar{f}_C(\mathbf{x}) = \frac{f_C(\mathbf{x}) - f_C^*}{|f_C^*| + \varepsilon_0}, \qquad \bar{f}_S(\zeta) = \frac{\zeta - f_S^*}{|f_S^*| + \varepsilon_0}. \tag{37}$$

For weights $\omega_1, \omega_2 \geq 0$ with $\omega_1 + \omega_2 = 1$, the augmented weighted Tchebycheff problem is

$$\min_{\mathbf{x}, \alpha_s, \mathbf{R}, \zeta} \max_{i \in \{1,2\}} \omega_i \left[\bar{f}_i + \xi(\bar{f}_C + \bar{f}_S)\right], \tag{38}$$

where $\xi > 0$. With epigraph variable α_T,

$$\min_{\mathbf{x}, \alpha_s, \mathbf{R}, \zeta, \alpha_T} \alpha_T \tag{39}$$

subject to

$$\omega_1[\bar{f}_C + \xi(\bar{f}_C + \bar{f}_S)] \leq \alpha_T, \tag{40}$$

$$\omega_2[\bar{f}_S + \xi(\bar{f}_C + \bar{f}_S)] \leq \alpha_T. \tag{41}$$

Different weights produce different Pareto trade-off points.

I. Final Deterministic Robust Problem

Before relaxation, the deterministic robust problem is

$$\min_{\mathbf{x},\mathbf{R},\alpha_s,\zeta,\alpha_T} \alpha_T \tag{42}$$

subject to the two Tchebycheff constraints, the CI constraints in (17)–(18),

$$\frac{1}{L}\sum_{l=1}^{L}\left|\mathrm{tr}(\mathbf{A}_l\mathbf{R})-\alpha_s d_l(\boldsymbol{\delta})\right|^2 \le \zeta, \quad \forall\boldsymbol{\delta}\in\mathcal{U}, \qquad \alpha_s \ge 0. \tag{43}$$

$$\mathrm{tr}(\mathbf{R}) \le P_{\max}, \qquad \mathbf{R}=\mathbf{x}\mathbf{x}^H, \qquad \mathbf{R}\succeq 0. \tag{44}$$

This problem bounds the beampattern error for every angle error in $\mathcal{U}$.

J. SDR and Waveform Recovery

The exact constraint $\mathbf{R}=\mathbf{x}\mathbf{x}^H$ is non-convex. To keep sensing and communication coupled, use the tighter SDR

$$\mathbf{R}\succeq\mathbf{x}\mathbf{x}^H. \tag{45}$$

By the Schur complement,

$$\begin{bmatrix}\mathbf{R} & \mathbf{x}\\ \mathbf{x}^H & 1\end{bmatrix}\succeq 0. \tag{46}$$

The relaxed constraints are

$$\mathrm{tr}(\mathbf{R}) \le P_{\max}, \qquad \begin{bmatrix}\mathbf{R} & \mathbf{x}\\ \mathbf{x}^H & 1\end{bmatrix}\succeq 0, \qquad \mathbf{R}\succeq 0. \tag{47}$$

After solving the SDP, SDR tightness is checked by

$$\rho_{\mathrm{rank}} = \frac{\lambda_1(\mathbf{R}^*)}{\sum_i \lambda_i(\mathbf{R}^*)}, \tag{48}$$

$$\rho_{\mathrm{res}} = \frac{\|\mathbf{R}^*-\mathbf{x}^*\mathbf{x}^{*H}\|_F}{\|\mathbf{R}^*\|_F}. \tag{49}$$

If $\rho_{\mathrm{rank}}\approx 1$ and $\rho_{\mathrm{res}}\approx 0$, the SDR is tight and $\mathbf{x}^*$ is used. Otherwise, Gaussian randomization draws

$$\widetilde{\mathbf{x}}\sim\mathcal{CN}(\mathbf{0},\mathbf{R}^*), \tag{50}$$

then scales and checks candidates. The feasible candidate with the best sensing objective is selected.

K. Simplified Simulation Problem

For a finite design set

$$\mathcal{D}=\{\boldsymbol{\delta}_1,\boldsymbol{\delta}_2,\ldots,\boldsymbol{\delta}_Q\}, \qquad \boldsymbol{\delta}_q=\mathbf{P}\mathbf{u}_q, \tag{51}$$

the full weighted Tchebycheff problem is still hard because $R_{\mathrm{sum}}(\mathbf{x})$ is not concave in $\mathbf{x}$. The simulation therefore removes the Tchebycheff constraints and keeps communication as fixed CI QoS. It uses the same scaled-desired-pattern form as the robust sensing formulation. The MATLAB problem is

$$\min_{\mathbf{x},\mathbf{R},\alpha_s,\tau} \tau \tag{52}$$

subject to

$$\frac{1}{L}\sum_{l=1}^{L}\left|\mathrm{tr}(\mathbf{A}_l\mathbf{R})-\alpha_s d_l(\boldsymbol{\delta}_q)\right|^2 \le \tau, \quad q=1,\ldots,Q, \qquad \alpha_s\ge 0, \quad \tau\ge 0. \tag{53}$$

$$\mathrm{Re}(\mathbf{h}_k^H\mathbf{x}s_k^*) \ge \sqrt{\Gamma_k\sigma^2}, \quad k=1,\ldots,K, \tag{54}$$

$$\mathrm{Im}(\mathbf{h}_k^H\mathbf{x}s_k^*) \le \left[\mathrm{Re}(\mathbf{h}_k^H\mathbf{x}s_k^*)-\sqrt{\Gamma_k\sigma^2}\right]\tan\frac{\pi}{M}, \tag{55}$$

$$-\mathrm{Im}(\mathbf{h}_k^H\mathbf{x}s_k^*) \le \left[\mathrm{Re}(\mathbf{h}_k^H\mathbf{x}s_k^*)-\sqrt{\Gamma_k\sigma^2}\right]\tan\frac{\pi}{M}, \tag{56}$$

for $k=1,\ldots,K$, and

$$\mathrm{tr}(\mathbf{R}) \le P_{\max}, \quad \begin{bmatrix}\mathbf{R} & \mathbf{x}\\ \mathbf{x}^H & 1\end{bmatrix}\succeq 0, \quad \mathbf{R}\succeq 0. \tag{57}$$

This is a sensing-oriented joint ISAC design with fixed communication QoS, not a sensing-only design.

L. Algorithm Flow

The simulation follows these steps.

1) Set N_t, K, N, M, $P_{\max}$, σ^2, Γ_k, nominal angles, grid, beam width, and $\mathbf{P}$.
2) Generate $\mathbf{H}$ and PSK symbol vector $\mathbf{s}$.
3) Construct $\mathbf{A}_l=\mathbf{a}(\theta_l)\mathbf{a}^H(\theta_l)$.
4) Use $\mathcal{D}_{\mathrm{nom}}=\{\mathbf{0}\}$ for the nominal baseline.
5) Use $\mathcal{D}_{\mathrm{rob}}=\{\boldsymbol{\delta}_q=\mathbf{P}\mathbf{u}_q, q=1,\ldots,Q\}$ for robust design, with boundary samples $\|\mathbf{u}_q\|_2=1$.
6) Build an independent test set $\mathcal{D}_{\mathrm{test}}=\{\boldsymbol{\delta}_q=\mathbf{P}\mathbf{u}_q, \|\mathbf{u}_q\|_2\le 1\}$.
7) Solve the nominal and robust SDPs.
8) Check SDR tightness. Use $\mathbf{x}^*$ directly if tight; otherwise use Gaussian randomization.
9) Evaluate sum rate, nominal MSE, average MSE, P95 MSE, worst MSE, CI feasibility, ρ_{rank}, ρ_{res}, and recovery method.

III. Simulation

A. Simulation Setup

All experiments use the same channel and communication symbols. The nominal and robust waveforms are evaluated on the same uncertainty test set.

Angle error is generated as $\boldsymbol{\delta}_q=\mathbf{P}\mathbf{u}_q$, where $\|\mathbf{u}_q\|_2\le 1$. In the radius sweep, $\mathbf{P}(\epsilon)=\mathrm{diag}(\epsilon, 0.6\epsilon)$ for $\epsilon=0^\circ,1^\circ,\ldots,7^\circ$. The same normalized samples are reused for all values of ϵ.

TABLE I
SIMULATION PARAMETERS

Item	Value
Transmit antennas	$N_t = 8$
Communication users	$K = 3$
Sensing targets	$N = 2$
Modulation	QPSK, $M = 4$
Power limit	$P_{\max} = 10$
Noise power	$\sigma^2 = 1$
CI SNR threshold	$\Gamma = 6$ dB
Nominal target angles	$[-20°, 25°]$
Angular grid	$-90° : 1° : 90°$, $L = 181$
Beam half width	$6°$
Fixed uncertainty	$\mathbf{P} = \mathrm{diag}(5°, 3°)$
Robust design samples	20 boundary samples
Test samples	3000 samples in the ball
Randomization	500 samples; unused because SDR is tight
Seeds	main=7, design=17, test=29

B. Evaluation Metrics

The actual beampattern is $p(\theta_l) = \mathrm{tr}(\mathbf{A}_l\mathbf{R})$. The evaluated sensing MSE is

$$\mathrm{MSE}(\boldsymbol{\delta}_q) = \frac{1}{L}\sum_{l=1}^{L} |p(\theta_l) - \alpha_s d_l(\boldsymbol{\delta}_q)|^2 . \tag{58}$$

The reported metrics are SumRate, NominalMSE at $\boldsymbol{\delta} = \mathbf{0}$, AverageMSE, P95MSE, WorstMSE, and CIFeasible. The MSE measures shape over the full angular grid. The scale α_s adjusts height, but cannot correct shifted peaks, uneven target-region response, or sidelobes.

C. Experiment 1: Robust vs. Nominal Joint ISAC

1) Goal and Procedure: This experiment fixes $\mathbf{P} = \mathrm{diag}(5°, 3°)$ and compares the nominal and robust designs. The nominal SDP uses only $\boldsymbol{\delta} = \mathbf{0}$. The robust SDP uses 20 design uncertainty samples. Both waveforms are tested on the same 3000 samples.

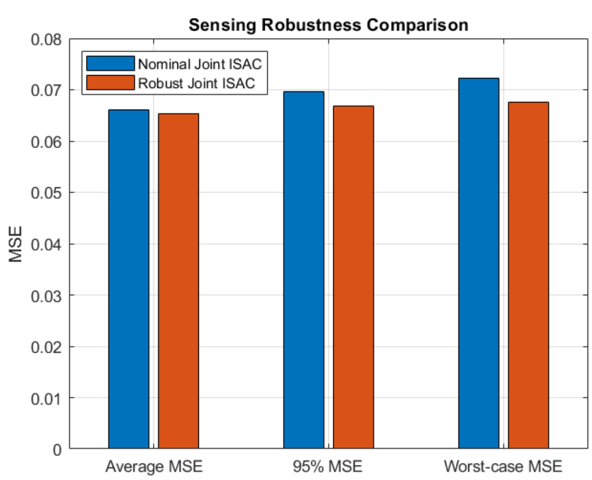

Fig. 1. Sensing robustness comparison under fixed angle uncertainty.

TABLE II
EXPERIMENT 1 RESULTS UNDER $\mathbf{P} = \mathrm{diag}(5°, 3°)$

Method	SumRate	NominalMSE	AvgMSE	P95MSE	WorstMSE	CI
Nominal Joint ISAC	7.6456	0.060602	0.066157	0.069569	0.072315	true
Robust Joint ISAC	7.5223	0.062395	0.065399	0.066878	0.067525	true

Method	ρ_{rank}	ρ_{res}	Recovery
Nominal Joint ISAC	1.0000	8.7217×10^{-9}	Direct optimized $\mathbf{x}$
Robust Joint ISAC	1.0000	3.6727×10^{-9}	Direct optimized $\mathbf{x}$

2) Discussion: The robust design reduces SumRate from 7.6456 to 7.5223, a 1.61% loss. It also increases no-error MSE from 0.060602 to 0.062395. This is expected because it is not optimized only for the nominal angles. Under uncertainty, it performs better: AverageMSE, P95MSE, and WorstMSE all decrease. WorstMSE drops from 0.072315 to 0.067525, a 6.62% improvement. The SDR check in Table II also shows $\rho_{\mathrm{rank}} = 1$ and $\rho_{\mathrm{res}} \approx 10^{-9}$, so the relaxation is tight and the optimized $\mathbf{x}$ is used directly. Thus robustness is gained by controlling high-risk sensing errors.

D. Experiment 2: Worst-Case MSE versus Uncertainty Radius

1) Goal and Procedure: This experiment uses $\mathbf{P}(\epsilon) = \mathrm{diag}(\epsilon, 0.6\epsilon)$ and sweeps $\epsilon = 0°$ to $7°$. The nominal waveform is fixed. The robust waveform is re-solved for each nonzero radius. The same normalized test samples are used for all values of ϵ.

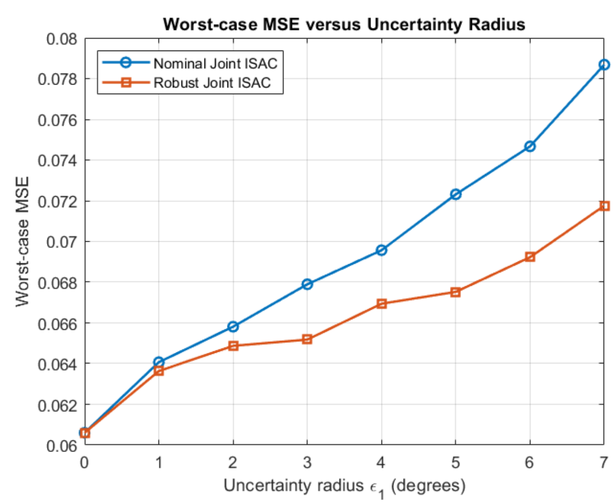

Fig. 2. Worst-case MSE versus uncertainty radius.

TABLE III
EXPERIMENT 2 RADIUS SWEEP RESULTS

ϵ_1	ϵ_2	Nom Worst	Rob Worst	Nom P95	Rob P95	Nom Avg	Rob Avg
0	0	0.060602	0.060602	0.060602	0.060602	0.060602	0.060602
1	0.6	0.064071	0.063635	0.064071	0.063635	0.063567	0.063515
2	1.2	0.065813	0.064873	0.064939	0.064647	0.063830	0.063860
3	1.8	0.067896	0.065180	0.066154	0.065016	0.064412	0.064408
4	2.4	0.069569	0.066942	0.067896	0.066350	0.065182	0.064834
5	3.0	0.072315	0.067525	0.069569	0.066878	0.066157	0.065399
6	3.6	0.074671	0.069228	0.072119	0.068524	0.067369	0.066206
7	4.2	0.078683	0.071756	0.074295	0.069837	0.068775	0.067378

2) Discussion: At $\epsilon = 0°$, both methods have the same WorstMSE, 0.060602. As the radius grows, both WorstMSE values increase, but the nominal curve grows faster. The robust design has lower WorstMSE for every nonzero radius. At $\epsilon = 7°$, WorstMSE decreases from 0.078683 to 0.071756, an 8.80% improvement. AverageMSE can be close; for example, at $\epsilon = 2°$, robust AverageMSE is slightly higher. This does not contradict the goal because the robust objective mainly controls high-risk and worst-case MSE.

E. Experiment 3: Beampattern Under Worst-Case Angle Error

1) Goal and Procedure: This experiment explains why the robust waveform is more stable. It searches the 3000 test samples from Experiment 1 and selects the sample that maximizes nominal MSE. The selected index is 375, with

$$\boldsymbol{\delta} = [-3.1953°, -2.2908°]. \tag{59}$$

The true target angles are

$$[-23.1953°, 22.7092°]. \tag{60}$$

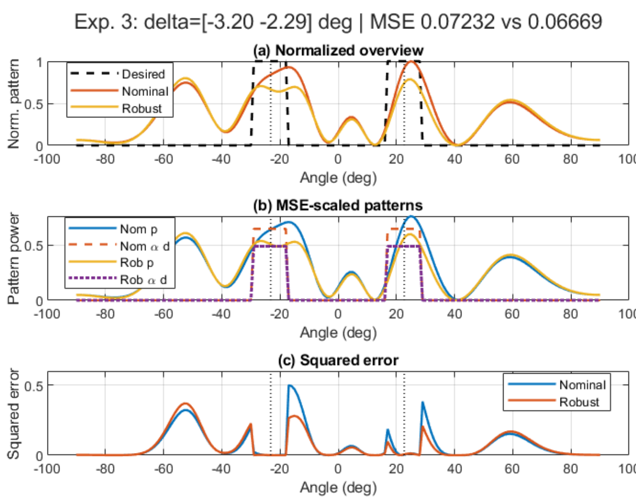

Fig. 3. Normalized overview, MSE-scaled patterns, and squared-error curves under the nominal worst-case angle error.

TABLE IV
EXPERIMENT 3 WORST-CASE SAMPLE

Item	Value
Worst index	375
$\boldsymbol{\delta}$	$[-3.195°, -2.291°]$
True target angles	$[-23.2°, 22.71°]$
Nominal MSE	0.072315
Robust MSE	0.066692

2) Discussion: Fig. 3(a) shows the normalized overview, which helps locate the target regions but is not the true MSE scale. Fig. 3(b) shows $p(\theta)$ and $\alpha_s d(\theta)$, the quantities used in MSE. The scale α_s adjusts the desired-pattern height. The remaining error comes from shifted peaks, uneven target-region response, and sidelobes. Fig. 3(c) directly plots $|p(\theta) - \alpha_s d(\theta)|^2$. The robust waveform has smaller error peaks, so MSE decreases from 0.072315 to 0.066692, a 7.78% improvement.

IV. Conclusion

The formulation and simulations show the same result. Nominal Joint ISAC gives better no-error performance, but it is sensitive to angle errors. Robust Joint ISAC uses angle-uncertainty samples during optimization. It slightly reduces SumRate and nominal MSE, but lowers high-risk sensing errors. In Experiment 1, robust design improves WorstMSE by 6.62% under $\mathbf{P} = \mathrm{diag}(5°, 3°)$. In Experiment 2, the gain reaches 8.80% at $\epsilon = 7°$. In Experiment 3, the squared-error curve confirms that the gain comes from better beampattern shape matching under shifted target angles, not from a larger peak magnitude.